Project Phase 2

Music Box

ENGR 3530 Electronics and instrumentation

Instructor: ***Hongbo Zhang***

All Team Member Names: ***David A. Williams***

Jaesung Kim

Jaden Jackson

Experiment Conducted on: ***04/14/2026***

Report Submitted on: ***04/28/2026***

Department of Engineering Technology

Middle Tennessee State University

Table of Contents

INTRODUCTION 3

Phase 1 3

Phase 2 4

METHODS and PROCEDURES 5

Phase 1 5

Calculations 6

Phase 2 7

Calculations 9

OBSERVATOINS, DATA, FINDINGS, and RESULTS 10

Phase 1 10

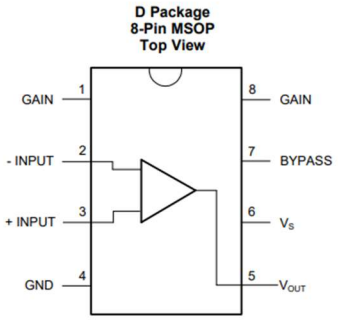
Phase 2 12

Discussions 14

INTRODUCTION

Phase 1

The primary objective of this project is to design and implement a functional, portable "Music Box" utilizing the LM386 low-voltage audio power amplifier. This project integrates several core engineering disciplines, including analog circuit design, hardware fabrication, and digital signal processing. The LM386 is a versatile integrated circuit specifically designed for battery-powered consumer applications, such as radio systems and portable speakers, due to its low quiescent current drain and ability to drive low-impedance loads with minimal external components. The specific pinout and functional descriptions for the LM386 are shown below in Figure 1.



Pin Functions

PIN		TYPE	DESCRIPTION
NAME	NO.		
GAIN	1	–	Gain setting pin
–INPUT	2	I	Inverting input
+INPUT	3	I	Noninverting input
GND	4	P	Ground reference
V_{OUT}	5	O	Output
V_S	6	P	Power supply voltage
BYPASS	7	O	Bypass decoupling path
GAIN	8	–	Gain setting pin

Figure 1. Pinout of LM386 chip

To verify the design and analyze the performance of the amplifier, several engineering tools are employed. Multisim is used for electronic schematic simulation to emulate the circuit's behavior in a virtual environment. For physical measurement, the NI MyDAQ serves as the data acquisition interface, bridging the physical breadboard circuit with the computer.

A key focus of the laboratory is the implementation of digital filters within LabVIEW. By applying low-pass, high-pass, and band-pass filters, we aim to manipulate the acquired sound signal to remove noise and observe the resulting changes in the power spectrum. This process demonstrates the fundamental principles of signal acquisition and the importance of filtering in real-world electronic systems.

Phase 2

The secondary objective of this project was to evolve the initial 'Music Box' design into a high-fidelity, two-way active speaker system. While Phase 1 utilized the LM386 low-voltage amplifier to drive a single full-range speaker, Phase 2 focuses on frequency specialization and power enhancement. By transitioning to the TDA2050 monolithic power amplifier, capable of delivering up to 32W, and a 14.8V Lithium-Ion battery system, this design provides significantly more headroom and dynamic range. A critical addition to this phase is the implementation of an active RC crossover network, which intelligently splits the audio signal into dedicated frequency paths for a Peerless 3.5” woofer and a Visaton 1” silk-dome tweeter. This multi-driver approach demonstrates advanced principles of analog signal processing and electro-acoustic integration.

METHODS and PROCEDURES

Phase 1

The implementation of the audio amplifier focused on a minimal-component configuration of the LM386 integrated circuit to evaluate its baseline performance. In this specific build, the gain control pins (1 and 8) and the bypass pin (7) were left open, which theoretically sets the internal gain to 20 V/V. The input signal was delivered to the non-inverting input (Pin 3) through a 1k ohm series resistor, while the inverting input (Pin 2) and ground pin (Pin 4) were tied directly to the common ground. The output stage (Pin 5) featured a 100 µF shunt capacitor to ground for high-frequency filtering and a 1000 µF capacitor going to the speaker. The circuit was powered by a 4V DC supply at Pin 6, stabilized by a parallel 0.47 µF decoupling capacitor. Performance was then characterized by measuring input and output parameters across a frequency range of 100 Hz to 5000 Hz.

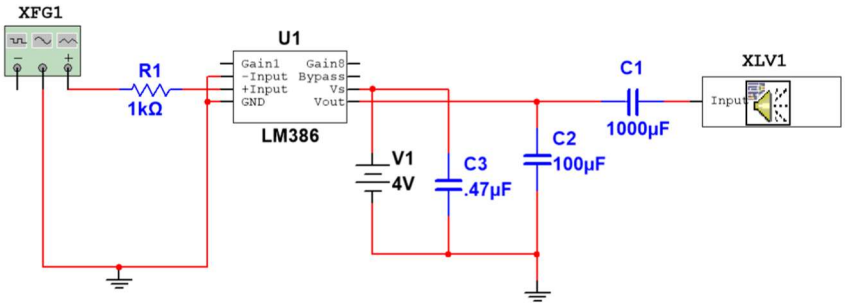


Figure 2. Multisim Circuit Layout

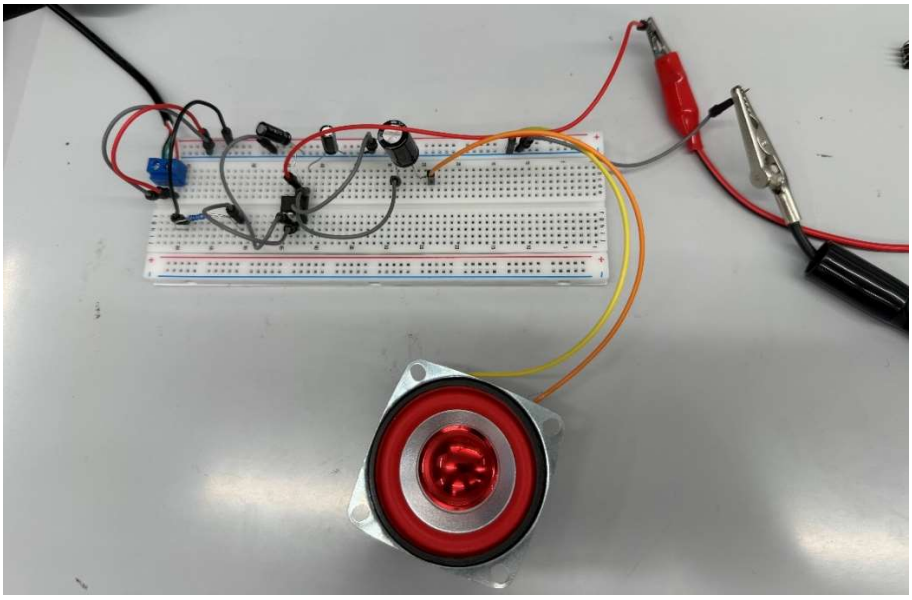


Figure 3. Physical Circuit Layout

Calculations

1. Voltage Gain (A_v): with pins 1 and 8 open, theoretical gain is 20.

$$A_v = \frac{V_{out}}{V_{in}} = \frac{0.73}{0.05} = 14.6\frac{V}{V}$$

$$20\log|A_v| = 20\log(14.6) = 23.3\ \text{dB}$$

2. Output Current (I_{out}):

$$I_{out} = \frac{V_{out}}{R_L} = \frac{0.73}{8} = 0.091\ \text{A} = 91\ \text{mA}$$

3. Current Gain (A_i):

$$A_i = \frac{I_{out}}{I_{in}} = \frac{91\ \text{mA}}{979 * 10^{-6}} = 93462\frac{\text{A}}{\text{A}}$$
$$20\log|A_i| = 20\log(93462) = 99.4\ \text{dB}$$

4. Load Power (P_L): The power delivered to the speaker.

$$P_L = V_{out} * I_{out} = 0.73\ \text{V} * .091\ \text{A} = 0.0668\ \text{W}\ (66.8\ \text{mW})$$

5. Input Power (P_i):

$$P_i = V_{in} * I_{in} = 0.05\ \text{V} * 979\ \mu\text{A} = 0.049\ \mu\text{W}$$

6. Power Gain (A_p) :

$$A_p = \frac{P_L}{P_i} = \frac{66.8\ \text{mW}}{0.049\ \mu\text{W}} = 1.36\ \text{M}\frac{\text{W}}{\text{W}}$$
$$10\log A_p = 10\log(1.36 * 10^6) = 61.35\ \text{dB}$$

7. DC Supply Power (P_{DC}) : The power consumed from the 4V source.

$$P_{DC} = V_s * I_s = 4\ \text{V} * 0.05\ \text{A} = 0.2\ \text{W}\ (200\ \text{mW})$$

8. Power Efficiency (eta) : The percentage of DC power converted to audio power.

$$\text{eta} = \left(\frac{P_L}{P_{DC}}\right) * 100\% = \frac{66.8\ \text{mW}}{200\ \text{mW}} * 100\% = 33.4\%$$

Phase 2

The implementation of the Phase 2 speaker system utilized a parallel-stage amplification architecture combined with pre-amplifier active filtering. The design process began with a Multisim simulation to verify the 3.4 kHz crossover point, achieved using 100Ω resistors and 0.47µF capacitors in a first-order RC configuration. To maintain signal integrity across both amplification stages, 100kΩ isolation resistors and 10µF input coupling capacitors were integrated into a shared 7.4V DC biasing network. The physical prototype was constructed on a solderless breadboard, where the TDA2050 chips were stabilized with high-capacity 1000µF output decoupling capacitors and local power supply bypass capacitors to mitigate voltage sag during low-frequency transients. Performance was characterized by measuring the distinct frequency roll-offs of the woofer and tweeter paths using an AC sweep and real-world audio signal analysis via an oscilloscope.

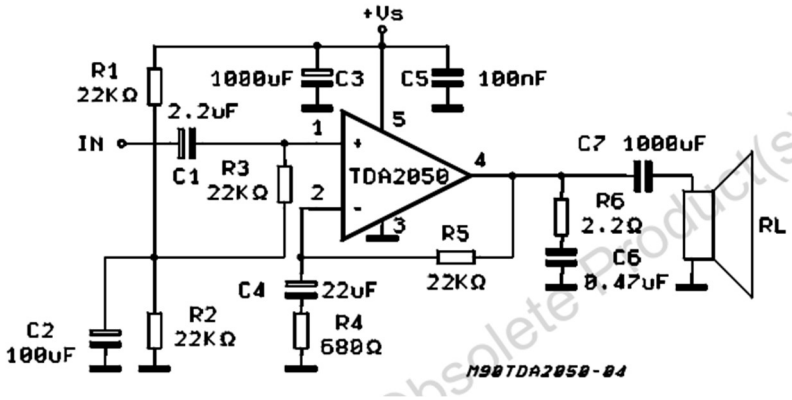


Figure 4. Provided Circuit Layout

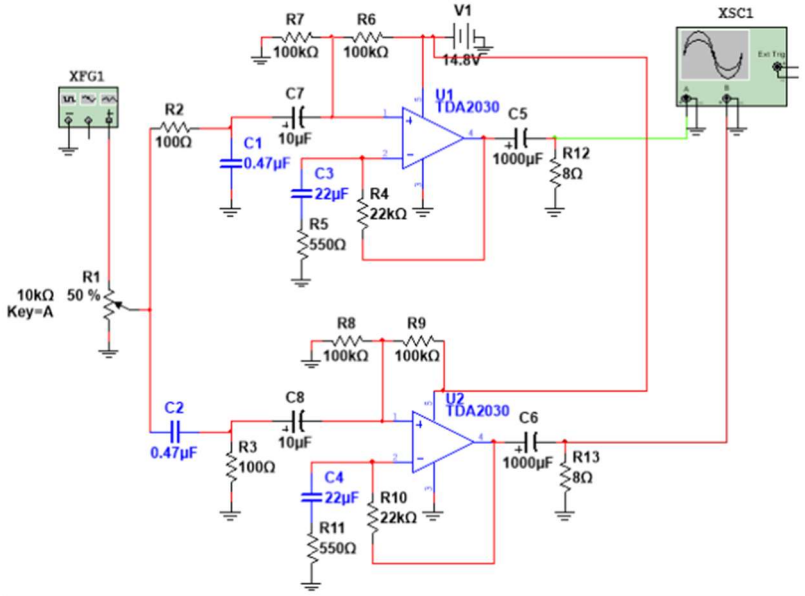


Figure 5. Multisim Circuit Layout

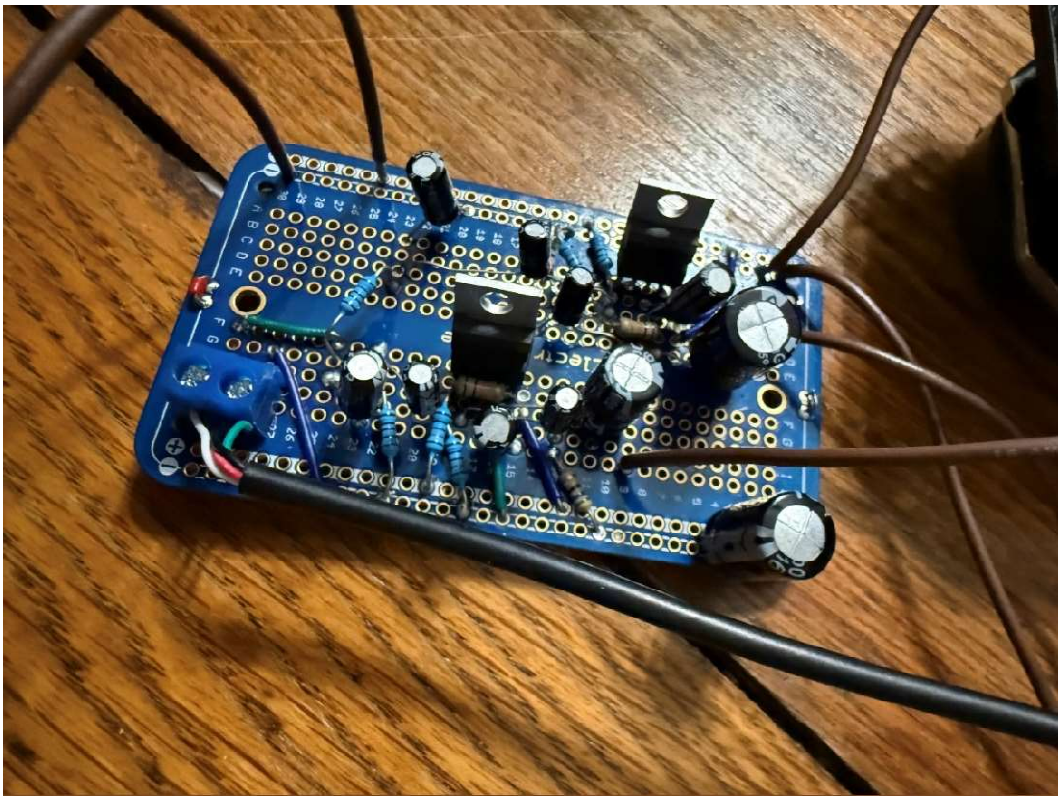


Figure 6. Physical Circuit Layout

Calculations

1. Voltage Gain (A_v): with pins 1 and 8 open, theoretical gain is 20.

$$A_v = 1 + \frac{R_f}{R_i} = 1 + \frac{22000}{550} = 41\frac{V}{V}$$

$$20\log|A_v| = 20\log(41) = 32.3\text{ dB}$$

2. Output Current (I_{out}):

$$I_{out} = \frac{V_{out}}{R_L} = \frac{3.8}{8} = 0.475\text{ A} = 475\text{ mA}$$

3. Current Gain (A_i):

$$A_i = \frac{I_{out}}{I_{in}} = \frac{475\text{ mA}}{0.1} = 4750\frac{A}{A}$$
$$20\log|A_i| = 20\log(4750) = 73.5\text{ dB}$$

4. Load Power (P_L): The power delivered to the speaker.

$$P_L = V_{out} * I_{out} = 3.8\text{ V} * .475\text{ A} = 1.81\text{ W}$$

5. Input Power (P_i):

$$P_i = V_{in} * I_{in} = 0.1\text{ V} * 100\ \mu\text{A} = 10\ \mu\text{W}$$

6. Power Gain (A_p) :

$$A_p = \frac{P_L}{P_i} = \frac{1810\text{ mW}}{0.01\ \mu\text{W}} = .181\text{ M}\frac{W}{W}$$
$$10\log A_p = 10\log(.181 * 10^6) = 52.5\text{ dB}$$

7. DC Supply Power (P_{DC}) : The power consumed from the 4V source.

$$P_{DC} = V_s * I_s = 14.8\text{ V} * 0.25\text{ A} = 3.7\text{ W}$$

8. Power Efficiency (eta) : The percentage of DC power converted to audio power.

$$\text{eta} = \left(\frac{P_L}{P_{DC}}\right) * 100\% = \frac{1.81\text{ mW}}{3.7\text{ mW}} * 100\% = 48.9\%$$

OBSERVATOINS, DATA, FINDINGS, and RESULTS

Phase 1

To verify the theoretical performance of our physical circuit, a simulation was conducted using NI Multisim. This simulation served as a controlled environment to establish a performance baseline, allowing us to observe the amplifier's behavior, specifically its voltage gain and frequency response, free from the environmental noise and component tolerances present in a physical breadboard. By comparing the ideal Multisim results with our empirical hardware measurements, we can better identify efficiency losses, clipping thresholds, and the impact of the specific output coupling capacitors used in our build.

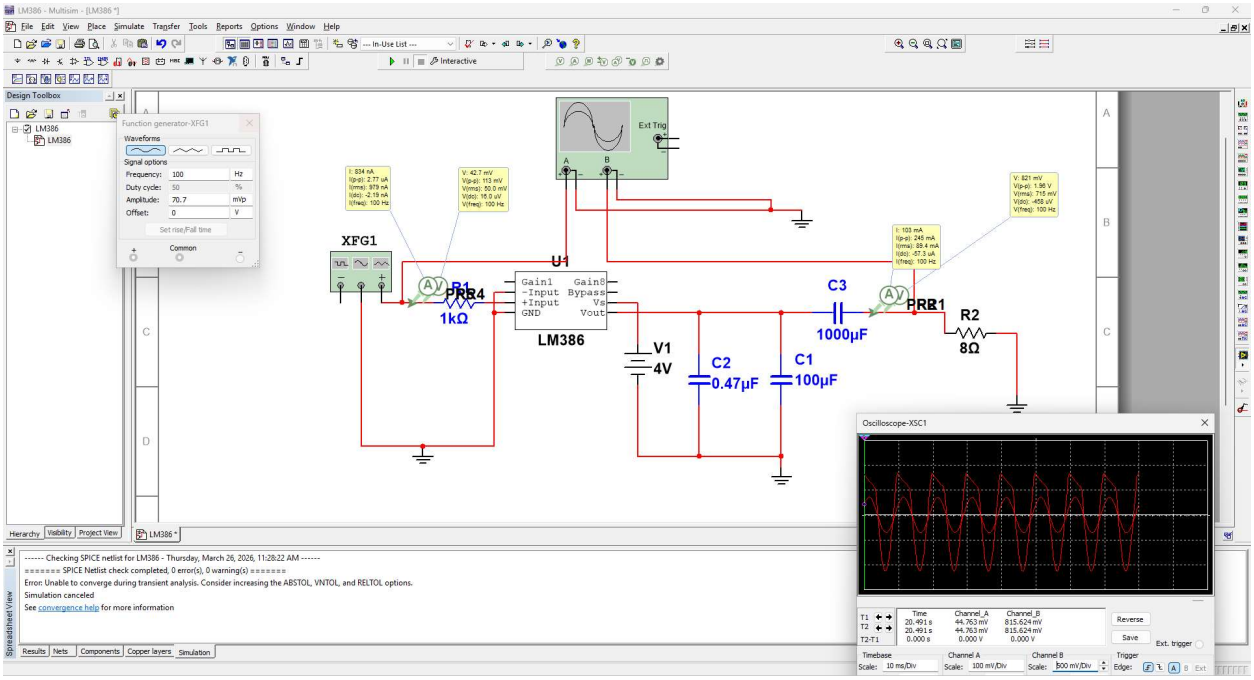


Figure 7. Multisim Simulation

Table 1. Measured values

Frequency (Hz)	Input Voltage (V) RMS	Input Current (μA)	Output Voltage (V) RMS	Output Current (mA)
100	0.050	0.979	0.715	89.4
500	0.050	0.979	0.729	91
1K	0.050	0.979	0.73	91.5
2k	0.050	0.979	0.737	92.2
4k	0.050	0.979	0.698	87.2
5k	0.050	0.979	0.661	82.4

The Multisim simulation was performed with a 50mVrms input and a 4V supply. While the theoretical calculations predicted a 1.0V output, the simulation showed a slight reduction to approximately 0.707 V due to the low supply voltage causing early signal saturation. Consequently, the measured output current through the 8Ω load was 88.4mA, which is lower than the calculated 125mA but accurately reflects the physical limitations of the 4V system.

Phase 2

The Phase 2 prototype demonstrated successful frequency separation and significant power improvement over the Phase 1 design. Simulation data from the AC Sweep confirmed a precise crossover intersection at 3.4 kHz, with the woofer magnitude rolling off effectively in the high-frequency range and the tweeter magnitude rising as intended.

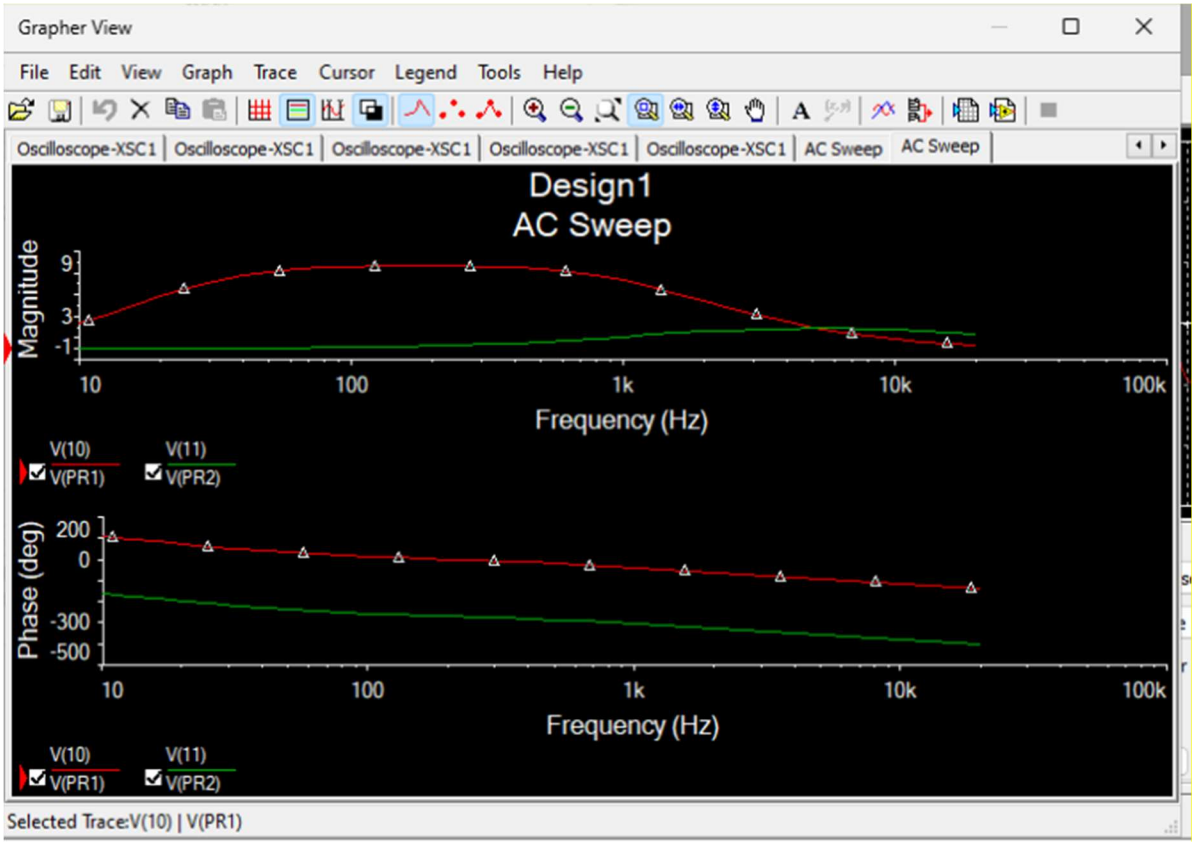


Figure 8. Multisim AC Sweep

Oscilloscope analysis of the physical breadboard circuit verified these findings, showing distinct amplitude dominance for the Peerless woofer at 100 Hz and the Visaton tweeter at 10 kHz. During physical testing, a baseline 60 Hz hum was observed while the signal source was connected to AC power; however, this was identified as ground-loop interference from the laptop charger and was eliminated by switching the source to battery power. Furthermore, while a slight bias drift was noted (9V/5V split) due to resistor tolerances, the TDA2050 chips maintained linear operation and provided clear, high-volume audio without observable clipping when driven at a 250mV peak input signal.

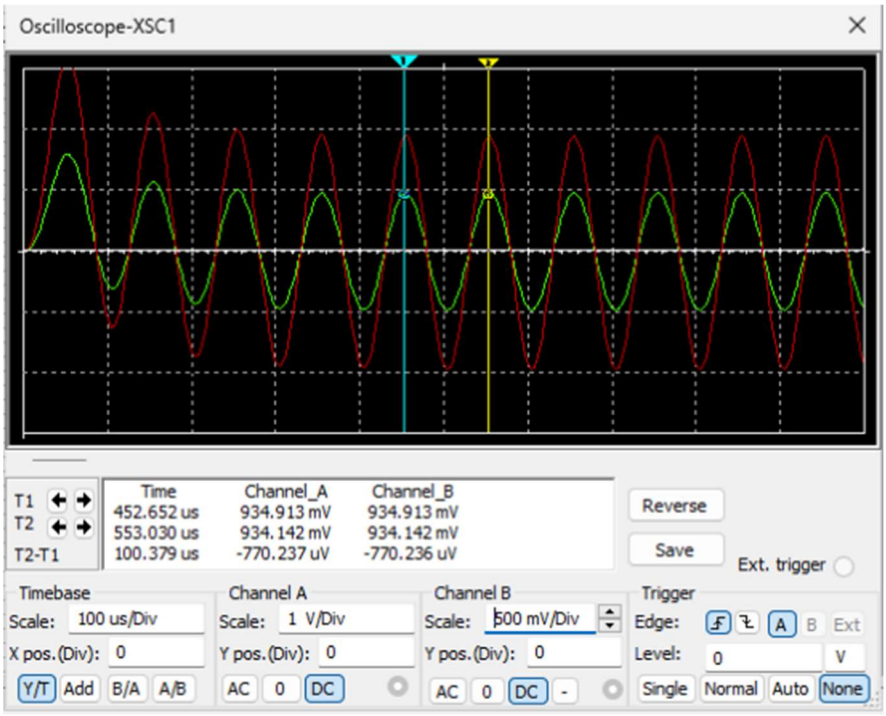


Figure 8. Multisim Oscilloscope

Discussions

In conclusion, the two-phase development of the 'Music Box' project successfully demonstrated the transition from basic signal amplification to advanced multi-driver acoustic engineering. While Phase 1 established a baseline for audio acquisition using the LM386, Phase 2 significantly elevated the system's performance by utilizing the TDA2050 power amplifier and a 14.8V high-headroom power supply. The successful implementation of an active RC crossover allowed for a sophisticated frequency division that optimized the output for the Peerless woofer and Visaton tweeter. Key engineering lessons were learned regarding the impact of resistor tolerances on DC biasing and the critical role of decoupling capacitors in maintaining stability on a solderless breadboard. Ultimately, the project fulfilled all technical objectives, resulting in a robust, portable, and high-fidelity speaker system that effectively bridges the gap between theoretical circuit design and practical, real-world application.